A project report on

PRIVACY PRESERVING AUTISM DISEASE PREDICTION WITH HOMOMORPHIC ENCRYPTION

ABSTRACT

Autism Spectrum Disorder (ASD) is a complex condition that affects how individuals communicate, interact, and behave. Early diagnosis is crucial because it allows for timely support and better long-term outcomes. In recent years, machine learning has become an important tool in healthcare, helping clinicians identify ASD by analyzing medical data in new ways. However, as these technologies handle sensitive patient information, protecting privacy has become a top priority. This project introduces a system for predicting autism that combines machine learning with a special kind of encryption called Homomorphic Encryption. With this approach, all patient data remains encrypted throughout the testing process, so personal details are never revealed, even during analysis. The model can make accurate predictions directly on encrypted data, ensuring privacy is not compromised. By adopting this method, we show that it is possible to build trustworthy AI tools for healthcare that both respect patient privacy and deliver strong performance. Our experiments confirm that the system maintains high accuracy while keeping data secure. This work demonstrates the promise of using advanced encryption techniques together with intelligent models to make healthcare safer and more reliable.

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LIST OF ACRONYMS

Acronym	Definition
ASD	Autism Spectrum Disorder
MRI	Magnetic Resonance Imaging
fMRI	Functional Magnetic Resonance Imaging
HE	Homomorphic Encryption
CKKS	Cheon-Kim-Kim-Song (Encryption Scheme)
FP	False Positive
CNN	Convolutional Neural Network
3D-CNN	Three-Dimensional Convolutional Neural Network
ABIDE	Autism Brain Imaging Data Exchange
NIfTI	Neuroimaging Informatics Technology Initiative
ML	Machine Learning
DL	Deep Learning
FHE	Fully Homomorphic Encryption
PHE	Partially Homomorphic Encryption
ReLU	Rectified Linear Unit
TP	True Positive
TN	True Negative

Chapter 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Autism Spectrum Disorder (ASD) is a complex and varied neurodevelopmental condition, impacting a person's ability to communicate, interact socially, and regulate behavior. Because its symptoms and severity differ so greatly among individuals, it is appropriately called a "spectrum" disorder. Early identification is critically important since timely intervention can significantly improve developmental outcomes and long-term quality of life. Despite this urgency, current diagnostic methods typically rely on behavioral assessments by expert specialists, which can be inherently subjective, demand significant resources, and may not be easily accessible to everyone.

Fortunately, breakthroughs in medical imaging, particularly Magnetic Resonance Imaging (MRI), have opened up new avenues for objective brain examination. MRI generates highly detailed, three-dimensional images that enable researchers to analyze subtle structural variations within the brain. This capability has naturally led to the adoption of machine learning to create automated tools for detecting ASD using neuroimaging data.

Deep learning techniques, especially three-dimensional Convolutional Neural Networks (3D CNNs), have proven highly effective at identifying complex spatial patterns in volumetric brain scans. However, implementing these powerful models in clinical settings raises serious concerns about data security. MRI scans contain sensitive patient health information and are strictly protected by privacy regulations. Storing or processing this data on shared cloud platforms or centralized servers creates unacceptable risks of unauthorized access and data breach.

To address these security challenges, there is a critical need for new privacy-preserving computational methods. Homomorphic Encryption (HE) stands out as a promising solution because it allows mathematical operations to be performed directly on encrypted data without ever needing to reveal the original, sensitive information. This ensures medical data remains fully protected throughout the entire analysis process.

Driven by this dual motivation, the need for accurate, automated diagnosis and the absolute requirement for data privacy this project introduces a hybrid framework combining deep learning and homomorphic encryption. Within this framework, a deep learning encoder is used to efficiently extract meaningful features from MRI scans, and a simple linear classifier then performs the prediction in a way that is compatible with encrypted computation. This combined approach allows for the secure and efficient detection of ASD while strictly upholding patient data privacy.

1.2 PROBLEM STATEMENT

The core goal of this research is to develop a reliable system capable of differentiating between individuals diagnosed with Autism Spectrum Disorder (ASD) and neurotypical individuals, based on their three-dimensional MRI scans, while rigorously protecting the confidentiality of all sensitive medical data during the entire prediction pipeline.

Achieving this objective requires overcoming several interconnected technical hurdles. First, raw MRI datasets often lack uniformity in resolution, size, and intensity, necessitating a robust preprocessing stage to convert the data into a standard format suitable for computational analysis. Second, the immense size and complexity (high dimensionality) of volumetric brain images demand a sophisticated and efficient method for feature extraction to identify the most relevant structural patterns.

Crucially, the main concern is guaranteeing data privacy during the model's inference phase. Standard machine learning models typically process unencrypted input, which can expose confidential patient information, especially when used in clinical or cloud-based environments. Although homomorphic encryption offers a mechanism to process protected data without decryption, it significantly constrains the types of mathematical operations that can be computed efficiently.

To reconcile these constraints, our proposed solution is structured as a two-stage process. Initially, a deep learning encoder is employed to generate compact, highly expressive feature representations from the raw MRI scans. These extracted features are then fed into a linear classifier, chosen specifically because its operations are highly amenable to encryption-based computation. This architectural design allows us to make secure predictions, benefiting from the robust feature learning capabilities of deep models while adhering to strict privacy requirements.

1.3 OBJECTIVES

The primary goals guiding this project are:

1. To design and implement a comprehensive preprocessing pipeline for efficiently loading, normalizing, and resizing three-dimensional MRI data stored in the NIfTI format.
2. To develop a robust deep learning-based encoder capable of extracting salient, meaningful features from volumetric MRI data.
3. To construct a simple linear classifier optimized for the binary differentiation of autistic and neurotypical subjects.
4. To successfully integrate homomorphic encryption into the inference phase, thereby enabling secure computation over the encrypted feature representations.
5. To rigorously evaluate the system's performance using established metrics, including accuracy and the consistency of predictions between plaintext and encrypted modes.
6. To compare the predictive outcomes generated by plaintext inference against those from encrypted inference to definitively confirm the system's correctness and reliability.

1.4 SCOPE OF THE WORK

This project is strictly focused on developing and assessing a privacy-preserving system for Autism Spectrum Disorder detection utilizing structural MRI data. The work relies on a defined dataset of brain MRI scans classified as either autistic or neurotypical. The model architecture includes a three-dimensional convolutional encoder for extracting features and a linear classifier for the final prediction.

Importantly, Homomorphic Encryption is applied exclusively to the classifier during the inference stage to ensure compatibility with encrypted operations. The training phase is conducted entirely in the plaintext domain to ensure computational efficiency and feasibility.

This research explicitly excludes: training models on encrypted data, implementing federated learning across multiple institutions, or deploying the system for real-time clinical use. The final system evaluation is performed offline using the designated dataset, and its ability to generalize to other datasets or different types of medical imaging is not within the scope of this study.

1.5 ORGANIZATION OF THE THESIS

The subsequent chapters of this report are structured as follows:

- Chapter 2 presents the literature review on ASD prediction, deep learning, and privacy-preserving machine learning techniques.
- Chapter 3 explains the proposed system design, including preprocessing, model architecture, and training process.
- Chapter 4 describes the implementation of homomorphic encryption and the encrypted inference process.
- Chapter 5 presents the experimental results, performance metrics, and comparison between plaintext and encrypted inference.
- Chapter 6 discusses the results, clinical relevance, computational overhead, and limitations of the proposed system.
- Chapter 7 summarizes the work, conclusions, and future research directions.
- Chapter 8 provides the source code and implementation details of the proposed system.

Chapter 2

LITERATURE REVIEW

2.1 Architectural Analysis of Privacy-Preserving Machine Learning Models

The current landscape of privacy-preserving machine learning in healthcare is rapidly evolving, driven by the need to balance predictive performance with strict data confidentiality. The following section provides a deeper, more analytical examination of each study, highlighting not just what was done, but how and why it matters.

The paper “Federated Anomaly Detection for Early Stage Diagnosis of Autism Spectrum Disorders using Serious Game Data (2024)”[1] introduces a novel intersection of behavioral analytics and secure distributed learning. The authors design a federated anomaly detection system where sensitive behavioral data captured through serious games is never centralized. Fully Homomorphic Encryption (FHE) ensures computations occur on encrypted data, preserving privacy at every stage. The FLAMENCO ASDTests dataset, which captures interaction patterns indicative of ASD traits, is used to train the model. Achieving an AP score of 74.22, the work demonstrates that early-stage ASD signals can be detected without compromising user data, though performance suggests room for richer feature representations.

In “A Privacy-Preserving Federated Learning Method with Homomorphic Encryption for Chronic Kidney Disease Stage Prediction (2025)” [2], the authors push the boundary of clinical prediction under encryption. They integrate Homomorphic Encryption into a Federated Learning pipeline, allowing multiple institutions to collaboratively train a model without exposing raw patient data. Using a Chronic Kidney Disease dataset, the model achieves 98.67% accuracy remarkably high given the encryption overhead. The key innovation lies in optimizing encrypted aggregation, reducing communication cost while maintaining precision, making it one of the most practically viable HE-FL systems.

The conceptual paper “Moving Beyond Traditional Data Protection: Homomorphic Encryption in Healthcare (2025)” [3] reframes privacy not as a constraint but as an architectural principle. Rather than focusing on a specific dataset, it proposes a generalized framework where healthcare analytics pipelines are designed natively around encrypted computation. It outlines system-level designs, threat models, and integration strategies for HE in clinical workflows. While lacking empirical validation, its value lies in shaping how future systems should be architected privacy-first, not privacy-added.

"PRISM: Privacy-Preserving Rare Disease Analysis using FHE" [4] addresses one of the most sensitive domains of genomic data. The authors employ fully homomorphic encryption to enable analysis of rare disease datasets without exposing genetic information. Given the uniqueness of genomic data, privacy risks are extremely high. PRISM focuses on enabling encrypted queries and statistical analysis over genomic sequences. Although no explicit accuracy metrics are reported, the contribution is foundational: proving that large-scale genomic analytics can be executed securely, albeit with computational trade-offs.

In "Privacy Preserving Machine Learning Framework for Medical Image Analysis using Quantized Fully Connected Neural Networks with TFHE (2025)" [5], the authors tackle the challenge of applying HE to image-based models. Traditional CNNs are computationally expensive under encryption, so they propose quantized fully connected networks compatible with TFHE. By simplifying model architecture and reducing precision, they enable encrypted inference on BreastMNIST and BloodMNIST datasets. The model achieved 88.2% accuracy on plain text and 87.5% on encrypted text.

The study "ASD Prediction using Machine Learning (2025)" [6] introduces a critical improvement: demographic awareness. Instead of treating all clients equally, the model stratifies participants by age groups (toddlers to adults), addressing bias and improving generalization. Using a stratified ASD dataset, the model achieves 95-98% accuracy. This work is significant because it shifts focus from just privacy to fairness within privacy-preserving systems, an often overlooked dimension.

In "Privacy Preservation and Identity Tracing Prevention in AI-Driven Eye Tracking for Interactive Learning Environments (2025)" [7], the authors explore behavioral biometrics, specifically eye-tracking data. The system is designed in two phases: first identifying vulnerabilities (e.g., identity leakage via gaze patterns), and then mitigating them using Federated Learning combined with secure identity management. Using eye-tracking datasets capturing fixation, saccades, and pupil dilation, the model achieves 89% accuracy. The work highlights that even subtle behavioral data requires strong privacy safeguards.

The paper "Privacy-Preserving Federated Vision Transformer with HE (2025)" [8] represents a shift toward modern architectures. By combining Vision Transformers (ViTs) with CKKS-based homomorphic encryption in a federated setting, the authors enable encrypted training of high-capacity models. Applied to multi-site lung cancer histopathology datasets, the system achieves 90.02% accuracy on encrypted data. This is a significant step forward, showing that even transformer-based models traditionally resource-heavy can operate in encrypted environments.

The study "HE-based Machine Learning CKD Prediction" [9] simplifies the pipeline by focusing on direct encryption of machine learning models. Instead of federated learning, the model itself is encrypted during training and inference. Using a CKD dataset, it achieves 91-93% accuracy on encrypted data. The strength lies in simplicity, but scalability across institutions remains limited compared to FL-based approaches.

In "Privacy-Preserving EMR Sharing using FHE and IOTA" [10], the authors extend beyond prediction into infrastructure. They propose a hybrid system combining blockchain (IOTA) with homomorphic encryption to enable secure sharing of Electronic Medical Records. The architecture ensures data integrity, traceability, and confidentiality. Tested on EMR datasets.

The paper "Privacy-Preserving Cancer Type Prediction with HE" [11] applies logistic regression directly over encrypted genomic data. Using a mutation dataset of 2713 patients, the model achieves 66-71% accuracy. The key challenge addressed is implementing nonlinear functions (like sigmoid) under encryption, which the authors approximate efficiently. This work reinforces that even classical models remain relevant when adapted for secure computation.

"Secure Logistic Regression Based on HE (2018)" [12] is one of the earlier foundational works in this space. It introduces encrypted gradient descent and polynomial approximation of activation functions to enable training under HE constraints. Using the Edinburgh medical dataset, it achieves 95% accuracy. Despite its age, it remains influential, forming the backbone for many later HE-based optimization techniques.

In “Enhanced Early Autism Screening with Distributed Facial Image Datasets and Deep Federated Learning (2025)” [13] , the authors leverage facial image analysis for ASD detection. Using domain-adaptive federated learning, the model learns from heterogeneous datasets (Kaggle and YouTube). Achieving 90% accuracy, the study demonstrates that visual biomarkers can be effectively used in decentralized, privacy-preserving environments.

The work “A Novel Black-Winged Kite Algorithm with Deep Learning for Autism Detection of Privacy Pressed Data (2025)” [14] introduces a bio-inspired optimization algorithm for feature selection. Combined with deep learning and privacy-preserving techniques, it improves model efficiency on sMRI datasets. With 85.42% accuracy, the contribution lies in optimization—reducing feature redundancy while maintaining privacy.

In “Federated Learning for Enhanced Privacy in Autism Prediction” (2025) [15], the authors combine FL with Differential Privacy, adding noise to prevent data leakage. Using Q-CHAT-10 toddler screening data, the model achieves 91.2% accuracy. This layered privacy approach (FL + DP) represents a more robust defense mechanism against inference attacks.

The study “Privacy-Preserving Deep Learning for Autism Spectrum Disorder Classification” [16] uses Secure Multi-Party Computation (SMPC), distributing data across multiple servers. Applied to ABIDE I fMRI and symptom logs, the model achieves 73.5% accuracy. While lower in performance, it offers stronger theoretical privacy guarantees compared to HE or FL alone.

“CryptoDL: Deep Neural Networks over Encrypted Data (2017)” [17] is a pioneering work enabling CNNs to operate on encrypted inputs. By replacing nonlinear activations with low-degree polynomials, the model becomes HE compatible. Tested on MNIST and CIFAR-10, it achieves 91.5% accuracy, proving that deep learning and encryption are not mutually exclusive.

In “PPFLHE: Privacy-Preserving Federated Learning Framework with Homomorphic Encryption (2025)” [18], the authors refine federated aggregation using CKKS encryption. Applied to APTOS 2019 and MosMedData, the model achieves 81.53% accuracy. The focus is on efficient encrypted communication, reducing bottlenecks in multi-client environments.

The paper “Explainable AI with Homomorphic Encryption for Secure Cloud-Based Diagnosis (2025)” [19] integrates explainability into encrypted models using SHAP. Applied to MIT-BIH behavioral signal data, it achieves 93.4% accuracy. This is critical moving beyond black-box encrypted models to interpretable secure AI.

Finally, “HeFUN: Homomorphic Encryption for Secure Neural Inference (2023)” [20] addresses one of the hardest problems: nonlinear activation functions under HE. The authors propose interactive protocols to approximate these functions efficiently. Using MNIST, the model achieves 99% accuracy, marking a major step toward practical encrypted deep learning.

2.2 Limitations of Existing Models

Despite the significant progress outlined in the literature, current privacy-preserving models face several critical limitations that hinder their widespread clinical adoption:

1. **Accuracy vs. Encryption Trade-off:** Cryptographic constraints inherent to Homomorphic Encryption (HE) require complex non-linear functions (like ReLU or sigmoid) to be replaced by simpler, less expressive polynomial approximations. This necessary simplification of the model architecture compromises its ability to learn intricate data patterns, often resulting in a decline in predictive accuracy compared to models run in plaintext.

2. **Massive Computational Overhead:** Operations performed on encrypted data are drastically slower ranging from 100 to 1000 times slower—and demand substantially more memory than standard, unencrypted calculations. This extreme computational overhead makes high-throughput or real-time prediction tasks difficult and limits the practical deployment of these systems.

3. **Limited Scalability in Real Systems:** While Federated Learning (FL) combined with HE is a theoretical solution for multi-institutional training,

real-world deployment struggles with scalability. Secure aggregation and communication of large encrypted models across many systems result in high communication costs and time delays, restricting most research to smaller, simpler experimental setups.

4. **Dataset Limitations:** A large portion of existing research relies on relatively small, simple datasets (e.g., tabular CKD data, MNIST images) or synthetic data. This limited evaluation scope, and the frequent failure to integrate complex, multimodal data (like imaging, behavior, and medical records), compromises the generalizability of these models to complex conditions like Autism Spectrum Disorder (ASD).

5. **Weak Handling of Complex Data (like ASD):** Accurate ASD diagnosis requires the analysis of complex spatial and temporal relationships in high-dimensional data, such as MRI scans. Current HE-compatible models often oversimplify the input and ignore these crucial relationships because the necessary complex models are too resource-intensive to run efficiently under encryption constraints.

6. **Poor Model Interpretability:** Encrypted models are frequently referred to as "black boxes" because the encryption protocols restrict the implementation of common explainability techniques (e.g., SHAP). In a clinical context, where medical professionals require a clear understanding of an AI-driven prediction for patient trust and safety, this lack of interpretability poses a major obstacle to adoption.

7. **Incomplete Privacy Guarantees:** Although labeled as privacy-preserving, these systems are not perfectly secure. Federated Learning, for instance, is known to be vulnerable to information leakage through shared gradients. Similarly, HE protects data during computation but does not defend against all possible inference attacks. Layered techniques, such as Differential Privacy, introduce noise for safety but often further degrade accuracy.

8. **Lack of End-to-End Systems:** Most studies focus narrowly on securing only one isolated phase, such as encrypted prediction or secure training aggregation. There is a notable absence of complete, end-to-end systems that

seamlessly manage the entire workflow from secure data ingestion through to final protected prediction, limiting their direct use in clinical practice.

9. **Optimization Inefficiency:** Training models with encryption (when implemented) is less efficient because the approximation of non-linear functions hinders the use of advanced, efficient optimization algorithms like Adam. This inefficiency leads to longer training times and potentially less accurate final model weights.

10. **No Real-Time or Edge Deployment:** Due to their massive memory footprint and computational latency, most HE-based models are too heavy and slow for lightweight, resource-constrained environments like mobile devices or edge computing systems. This prevents their application in real-time screening or continuous monitoring scenarios crucial for early ASD detection.

2.3 SUMMARY

Table 2.1:Literature Review summary

Sl No.	Title and year of publication	Methodology	Dataset	Disadvantages	Accuracy/performance
1	Federated Anomaly Detection for Early stage diagnosis of Autism Spectrum Disorders using Serious Game Data [2024]	Federated Learning + Anomaly Detection	Serious game dataset	High communication cost, slow convergence	Ap score=74.22
2	A Privacy-Preserving Federated Learning Method with Homomorphic Encryption for Chronic Kidney Disease Stage Prediction [2025]	Federated Learning + Homomorphic Encryption	Clinical Chronic Kidney Disease (CKD)	Heavy computation due to encryption	98%
3	Moving Beyond Traditional Data Protection: Homomorphic Encryption in Healthcare [2025]	Conceptual healthcare privacy framework using HE for secure analytics	Conceptual Idea	Very slow processing speed	Homomorphic encryption allows secure data sharing for AI, improving accuracy across multiple

					instituti ons.
4	PRISM: Privacy-preser ving rare disease analysis using FHE [2025]	Fully Homomorphic Encryption used for encrypted rare-disease genomic analytics	Rare disease genomic datasets	High memory and processing requirements	Correctl y finds the disease gene TMCO1 with fast and secure processi ng.
5	Privacy preserving machine learning framework for medical image analysis using quantized fully connected neural networks with TFHE based inference[202 5]	Fully Connected Neural Network (FCNN) with Quantization-Aw are Training (QAT) and TFHE-based encrypted inference.	Medical image dataset(mnist)	High computational cost and slow inference due to TFHE	88.2%(p laintext) 87.5% (encrypt ed data)
6	ASD Prediction using ML [2025]	Stratified data splits + local SVM/LR on each site + FL-style aggregation into a global meta-classifier.	Stratified demographic ASD dataset	Limited model complexity, data scarcity, high communicatio n cost, and client heterogeneity issues.	85–90%
7	Privacy Preservation and Identity Tracing Prevention in AI-Driven Eye Tracking for Interactive	Phase 1 demonstrates backtracking vulnerability; Phase 2 utilizes Federated Learning (FL) with a secure	Serious-game -based eye-tracking data (gaze patterns: fixation, saccades,	Technically strong but computational ly heavy, with ethical concerns and limited real-world	89%

	Learning Environments [2025]	identity management system	pupil dilation) from students.	generalization.	
8	Privacy-Preserving Federated Vision Transformer with HE (2025)	Vision Transformer + FL + HE	Multi-site lung cancer histopathology image dataset used across three federated clients.	high computational cost, accuracy drop, and limited deployment	90.02% (encrypted) 96.12% (plaintext).
9	HE-based Machine Learning CKD Prediction [2025]	CKKS-based Homomorphic Encryption with Federated Learning, where hospitals encrypt clinical data and perform secure inference using Logistic Regression/MLP models.	UCI CKD dataset and multi-hospital clinical data	High cost, slow inference, limited models, large ciphertexts, and complex key management.	91–93% (encrypted) 94–96% (plaintext).
10	Privacy-Preserving EMR Sharing using FHE, IOTA [2024]	CKKS-based FHE for EMR encryption combined with IOTA Tangle + MAM for secure, decentralized patient-doctor data sharing.	Synthetic/real EMR dataset	high computation cost, large ciphertext size, key management complexity, scalability issues, and limited real-time usage.	100% data confidentiality and integrity
11	Privacy-preserving cancer type prediction with HE [2023]	Logistic Regression with BFV Homomorphic Encryption for secure cancer type prediction on encrypted genomic data.	TCGA Pan-Cancer dataset (2,713 patients, 10 cancer types, SNVs & CNVs).	high computation time, limited to simple models, large data size, and complex key management.	66–71% accuracy, AUC up to 0.94(same in encrypted)

12	Secure Logistic Regression Based on HE[2018]	Logistic regression trained using encrypted gradient descent with polynomial sigmoid approximation	Synthetic/Edinburg medical dataset	High training time, slight precision loss, limited model complexity, and poor scalability.	~95% (plaintext) slightly drop on encrypted
13	Enhanced Early Autism Screening with Distributed Facial Image Datasets and Deep Federated Learning [2025]	Xception CNN with Federated Learning (FedAvg) for distributed facial image-based ASD detection with domain adaptation.	Kaggle ASD facial dataset (~2000 images) + YTUIA institutional dataset.	high communication cost, heavy model, data quality issues, and ethical concerns.	90%
14	A Novel Black-Winged Kite Algorithm with Deep Learning for Autism Detection of Privacy Preserved Data (2025)	Black-Winged Kite Algorithm (BKA) for feature selection + Deep Neural Network with Paillier Homomorphic Encryption.	UCI ASD dataset	high computational cost, unproven optimization algorithm, limited to tabular data, and deployment complexity.	95.42%
15	Federated Learning for Enhanced Privacy in Autism Prediction [2025]	Federated Learning (FedAvg) with CNN ensemble for ASD detection using distributed facial and behavioral data.	Q-Multi-source ASD data	high communication cost, non-IID data issues, high hardware requirements, and ethical concerns.	91.2%
16	Privacy-preserving Deep Learning for Autism Spectrum Disorder	Federated Hypergraph Neural Network (HGNN) with multimodal fusion (fMRI + phenotypic data)	ABIDE I (fMRI), Patient Symptom Logs	high communication cost, performance drop vs centralized models,	73.5%

	Classification [2024]	using encrypted parameter sharing (FedAvg)		complex model, and data heterogeneity.	
17	CRYPTODL: Deep neural networks over encrypted networks 2025	CNN with low degree polynomial activation function	Tested on MNIST and CIFAR-10,	high computation cost, large ciphertext size, data expansion issues	91.5%accuracy
18	PPFLHE: Privacy-Preserving Federated Learning Framework with Homomorphic Encryption	FedAvg with Paillier Homomorphic encryption and ack mechanism	APTOS 2019 and CIFAR-10	Computational overhead and cannot be suitable for explosive realtime data	81.53% classification accuracy
19	Explainable AI with Homomorphic Encryption for Secure Cloud-Based Diagnosis	integrates explainability into encrypted models using SHAP	MIT-BIH Arrhythmia Database	Computationally expensive,approximation errors explained by 0.8% drop in accuracy between plain and encrypted	93.4% accuracy on encrypted data,94.2% on plain text.
20	HeFUN: Homomorphic Encryption for Secure Neural Inference (2023)	HeFUN, a framework based on a new paradigm called interactive Leveled Homomorphic Encryption (iLHE)	MNIST dataset of 70000 handwritten digits	The "iLHE" paradigm requires interaction between the client and the server.	99% accuracy

Chapter 3

SYSTEM DESIGN

3.1 OVERALL SYSTEM PIPELINE

This research proposes a unified system pipeline that integrates dataset preparation and model development for Autism Spectrum Disorder (ASD) classification based on MRI data. The source data originates from a preprocessed, combined subset of the Autism Brain Imaging Data Exchange (ABIDE 1 & ABIDE 2), a collection of large-scale brain MRI scans gathered across multiple research institutions. This dataset, comprising approximately 2,221 3D MRI volumes, is categorized into two classes: Autistic (Label = 1) and Typical Control (Label = 0). Each volume is stored in the NIfTI (.nii/.nii.gz) format and contains voxel-level intensity values.

The system functions in two distinct phases: a conventional training phase (plaintext) and a privacy-preserving inference phase (encrypted). During training, the MRI volumes are loaded, preprocessed, and fed into a deep learning model. For secure inference, the model extracts features which are then encrypted using the CKKS homomorphic encryption scheme, allowing for secure classification without revealing any sensitive patient data.

3.2 DATA ORGANIZATION

The dataset is organized into two primary folders:

- Autistic
- Typical_Control

A custom PyTorch dataset class named MRIDataset is used for loading MRI data. The dataset loader scans all directories, identifies .nii and .nii.gz files, assigns labels, and stores file paths for efficient access during training and testing.

The NiBabel library is used to read NIfTI MRI files and convert them into multidimensional NumPy arrays representing volumetric brain scans.

To ensure balanced learning, equal samples are selected from both classes. The number of samples per class is defined as:

n=min(Autistic Samples,Control Samples,750)

A total of 750 MRI scans are selected from each category, resulting in:

- Training Set: 1500 samples
- Testing Set: Remaining ~721 samples

The dataset is randomly shuffled before splitting to ensure unbiased distribution.

3.3 DATA PREPROCESSING

Given the inherent variations in resolution, orientation, and intensity that exist across different MRI acquisitions, a robust preprocessing pipeline is essential. This pipeline standardizes all input samples to a uniform format before they are processed by the deep learning model.

3.3.1 INTENSITY NORMALIZATION

To ensure consistency across MRI scans, each volume is normalized to the range [0, 1] using

$$x' = \frac{x - x_{min}}{x_{max} - x_{min} + \epsilon}$$

where $\epsilon=10^{-5}$ prevents division by zero.

This critical standardization step promotes stable model training and mitigates the impact of wide intensity variations on overall performance.

3.3.2 DIMENSIONALITY HANDLING

The preprocessing stage handles dimensionality inconsistencies in the MRI data as follows:

- **4D volumes** : reduced to 3D by selecting the first slice along the fourth dimension

- **2D images** : expanded to a 3D format by replicating the slice along a new axis.
- **3D volumes** : utilized directly.

This procedure guarantees that every input is converted to a consistent three-dimensional tensor format required for the 3D Convolutional Neural Network.

3.3.3 SPATIAL RESAMPLING

All MRI volumes are resized to a fixed resolution of : 48×48×48 using trilinear interpolation. This ensures that all inputs have the same spatial dimensions, which is required for batch processing in neural networks.

The selected resolution provides a balance between computational efficiency and preservation of structural information.

3.3.4 DATA AUGMENTATION

Data augmentation is dynamically implemented during the training phase to enhance model generalization and mitigate overfitting. The augmentation techniques applied include:

- Random flip along the sagittal axis (axis 0)
- Random flip along the coronal axis (axis 1)

Each of these spatial transformations is applied with a probability of 0.5. These augmentations effectively increase data variability while scrupulously preserving the fundamental anatomical structure and meaning of the volumetric data.

3.4 MODEL ARCHITECTURE OVERVIEW

The deep learning architecture employed in this project is a three-dimensional Convolutional Autoencoder augmented with a dedicated classification head. This architecture is composed of three primary modules:

1. **Encoder** : Responsible for extracting compact and meaningful feature representations from the input MRI volumes.

2. **Decoder** : Responsible for attempting to reconstruct the original input volume from the extracted features.
3. **Classifier** ; A final linear layer that performs the binary classification of the subject.

The training process utilizes a multi-objective optimization framework that simultaneously balances the reconstruction and classification tasks. This dual objective is crucial because the reconstruction goal serves as a vital regularization mechanism. It forces the model to not only learn features discriminative for classification but also to maintain the integrity of the underlying structural information in the MRI data, thus enhancing generalization and preventing overfitting.

3.5 ENCODER ARCHITECTURE

The Encoder is responsible for extracting meaningful features from the input MRI volume. It takes an input tensor of shape (batch_size, 1, 48, 48, 48).

The Encoder consists of three convolutional blocks, each followed by a max-pooling layer:

- Conv Block 1:
  32 filters, kernel size 3, ReLU activation
  Output size: 24 × 24 × 24
- Conv Block 2:
  64 filters, kernel size 3, ReLU activation
  Output size: 12 × 12 × 12
- Conv Block 3:
  128 filters, kernel size 3, ReLU activation
  Output size: 6 × 6 × 6

After convolution, the feature maps are flattened into a vector of size 27,648 which is passed through a fully connected layer to produce a 128-dimensional latent vector.

Table 3.1: Encoder Architecture

Layer	Operation	Output	Activation
FC	Linear	27,648	—
Reshape	—	128 X 6 X 6 X 6	—
Deconv1	ConvTranspose3D	64 X 12 X 12 X 12	ReLU
Deconv1	ConvTranspose3D	32 X 24 X 24 X 24	
Deconv1	ConvTranspose3D	1 X 48 X 48 X 48	ReLu

Interpretation of Table 3.1: Encoder Architecture

The architecture shown in this table describes how the model reconstructs a 3D MRI image from compressed features. First, a fully connected layer processes a flattened feature vector of size 27,648. This vector is then reshaped into a 3D structure of size $128 \times 6 \times 6 \times 6$.

After that, a series of transposed convolution (deconvolution) layers are used to gradually increase the spatial size of the data. The first layer expands it to $64 \times 12 \times 12 \times 12$, followed by another layer that increases it to $32 \times 24 \times 24 \times 24$. Finally, the last layer reconstructs the original MRI size of $1 \times 48 \times 48 \times 48$.

ReLU activation is applied in these layers to help the model learn complex patterns. Overall, this architecture helps in rebuilding the original MRI image from a compressed representation.

3.6 CLASSIFIER ARCHITECTURE

The Classifier is a single linear layer that maps the 128-dimensional latent vector to a two-dimensional output vector (logits) representing the classes:

- Autistic (1)
- Typical Control (0)

This design is intentionally simple to ensure compatibility with homomorphic encryption. Homomorphic encryption supports efficient computation of linear operations such as matrix multiplication and addition, but nonlinear operations are computationally expensive. Therefore, only the Classifier operates in the encrypted domain.

During training, the logits are passed to the cross-entropy loss function. During inference, the predicted class is determined by selecting the index with the highest score. The encrypted inference process preserves the prediction outcome, ensuring that predictions from encrypted data match those from plaintext computation.

3.7 PREPROCESSING AND AUGMENTATION PIPELINE

The comprehensive preprocessing and augmentation pipeline is applied dynamically as data is loaded for training. The sequential steps involved are:

Steps include:

1. Loading the raw MRI data utilizing the NiBabel library.
2. Normalizing intensity values to the range [0, 1].
3. Converting any 4D data volumes to 3D.
4. Expanding any 2D images to a 3D format.
5. Resizing all volumes to the fixed spatial resolution of 48 x 48 x 48
6. Applying random flips for data augmentation (during training only).
7. Converting the final processed data into PyTorch tensor format.

This standardized pipeline is essential for ensuring data consistency and significantly contributing to the model's ability to generalize to unseen samples.

3.8 LOSS FUNCTION AND OPTIMIZATION

The training process is driven by a composite objective that strategically combines two distinct loss functions:

- **Classification Loss:** Standard Cross-entropy loss, which is used to ensure the accuracy of the binary prediction.
- **Reconstruction Loss:** Mean Squared Error (MSE) loss, which is used to measure the quality of the image reconstruction.

The total loss is defined as:

$$\mathbf{L = L_{classification} + 0.2 \times L_{reconstruction}}$$

While the classification component focuses on discriminative power, the weighted reconstruction loss functions effectively as a regularizer, encouraging the latent features to retain meaningful structural information. Optimization is handled by the Adam optimizer, utilizing a fixed learning rate of 1e-4.

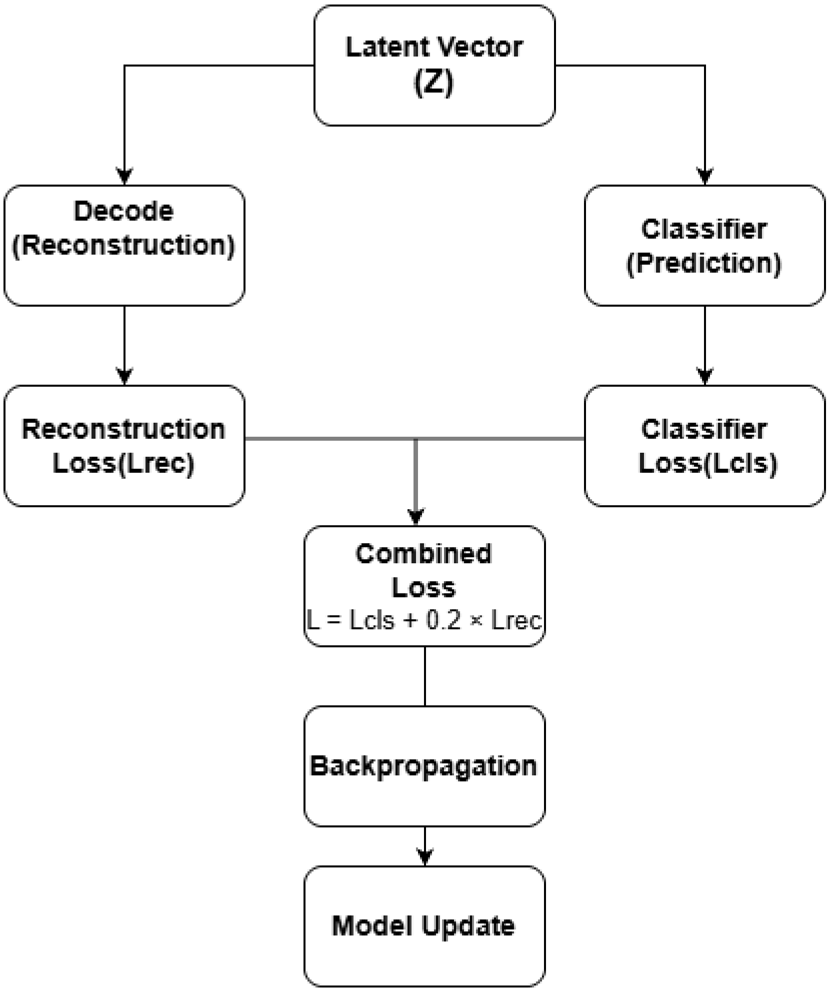

Figure 3.1: Training Pipeline of the Proposed Model

3.9 TRAINING CONFIGURATION

Table 3.2: Training Parameters

Parameter	**Value**
Optimizer	Adam
Learning Rate	1e-4
Batch Size	2
Epochs	50
Loss Function	Cross-Entropy + MSE
Reconstruction Weight	0.2
Input Size	48 x 48 x48
Latent Dimension	128

Interpretation of Table 3.2: Training Configuration

This table shows the main settings used to train the model. The Adam optimizer with a learning rate of 1e-4 is used for stable learning. A batch size of 2 is chosen due to the large size of 3D MRI data, and the model is trained for 50 epochs.

A combination of Cross-Entropy and MSE loss is used to handle both classification and reconstruction, with a reconstruction weight of 0.2. The input size is 48 × 48 × 48, and the data is compressed into a latent dimension of 128 for efficient feature learning.

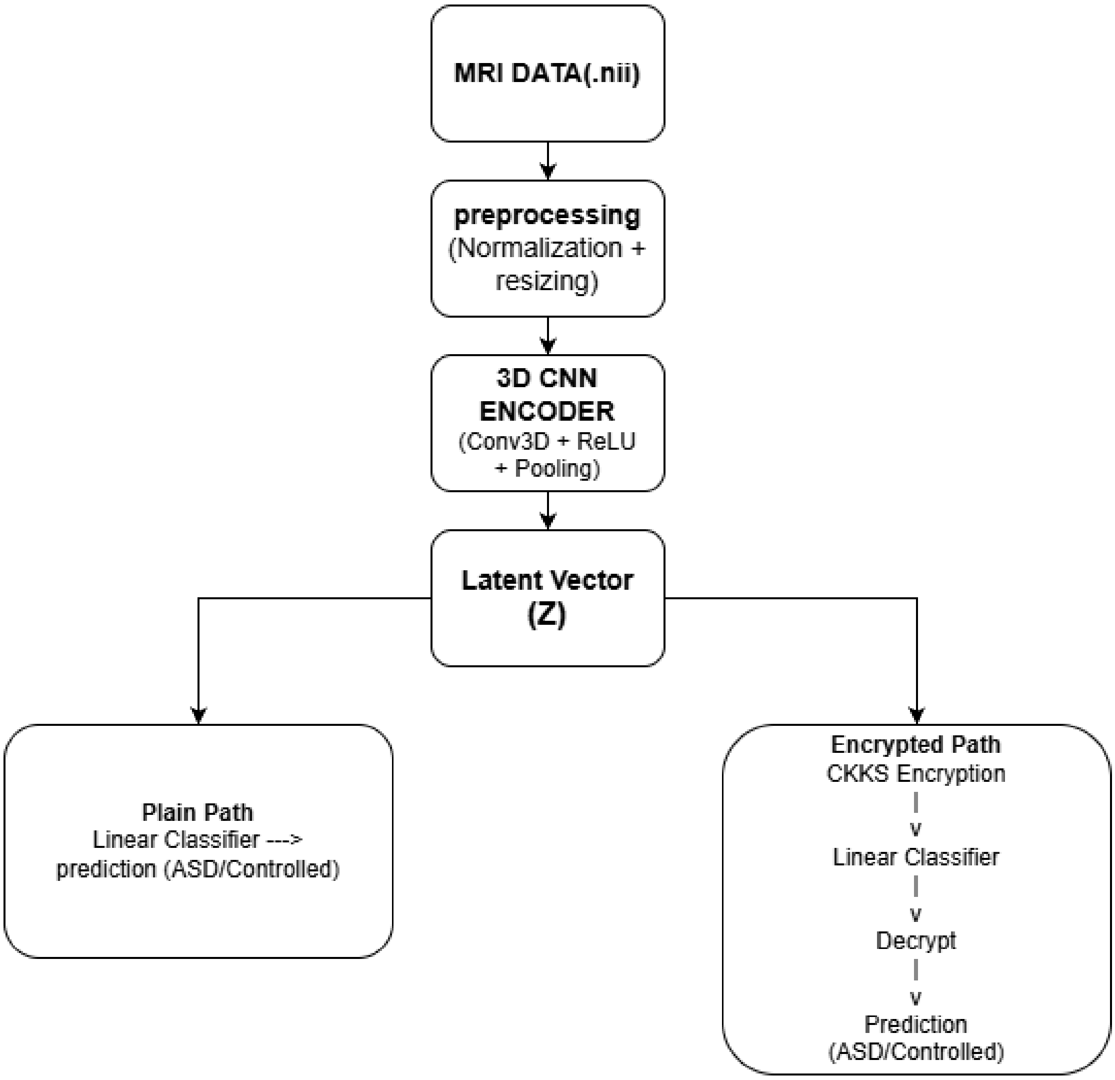

Figure 3.2: Privacy-Preserving Inference Pipeline using CKKS

CHAPTER 4

HOMOMORPHIC ENCRYPTION FOR SECURE INFERENCE

4.1 OVERVIEW OF HOMOMORPHIC ENCRYPTION

Homomorphic Encryption (HE) is a specialized cryptographic technique that enables arbitrary mathematical computations to be performed directly on encrypted data without ever requiring the data to be decrypted. This capability is fundamentally important because it allows sensitive information to remain fully protected throughout the entire computation process, even when executed on an untrusted server. In an HE system, data is encrypted using a public key, processed in its ciphertext form, and the resulting encrypted output, when decrypted with the corresponding private key, is exactly the same as if the operations had been performed on the original plaintext.

HE schemes fundamentally support linear operations, specifically addition and multiplication. These operations form the necessary building blocks for implementing linear machine learning models within the secure, encrypted domain. Unlike conventional encryption, HE facilitates the secure delegation of computation to third-party or cloud environments.

In practical machine learning applications, Fully Homomorphic Encryption (FHE), which supports unlimited operations, is often too resource-intensive. Consequently, leveled homomorphic encryption (LHE) schemes which support a sufficient but limited number of operations before ciphertext "noise" grows too large are utilized. This project adopts an LHE approach to ensure efficient and practical encrypted inference.

4.2 THE CKKS ENCRYPTION SCHEME

The CKKS (Cheon-Kim-Kim-Song) scheme is a contemporary and widely adopted homomorphic encryption method particularly optimized for approximate arithmetic on real-valued, floating-point data. This makes it ideally suited for machine learning, where the need for absolute mathematical precision is secondary to computational efficiency, and small numerical approximations are permissible without affecting the model's final predictive outcome.

CKKS is highly efficient for vector-based computations common in deep learning, natively supporting operations such as vector addition, multiplication, and, crucially, matrix-vector multiplication. It also supports batching, which allows multiple data points (or vector elements) to be encoded and processed simultaneously within a single ciphertext, significantly improving overall throughput. The Classifier in this work operates on floating-point feature vectors, making CKKS the preferred choice for maintaining both accuracy and efficiency in the encrypted domain.

4.3 TENSEAL LIBRARY AND CONTEXT SETUP

The implementation of all homomorphic encryption operations in this project is managed by the TenSEAL library. TenSEAL serves as a user-friendly, high-level interface built upon the robust Microsoft SEAL library, providing excellent support for the CKKS encryption scheme.

The security and performance parameters are meticulously defined within the encryption context setup:

- Scheme: CKKS (optimized for real numbers)
- Polynomial Modulus Degree: 8192
- Coefficient Modulus: [60, 40, 40, 40, 60]
- Global Scale: 2^40
- Security Level: 128-bit
- Galois Keys: Enabled

The polynomial modulus degree controls both the security strength and the associated computational overhead. The coefficient modulus specifies the maximum number of sequential multiplicative operations that can be performed before the ciphertext noise inevitably degrades the result. The global scale is set to maintain the necessary numerical precision for the approximate arithmetic. Galois keys are generated specifically to enable vector rotation operations, which are essential for efficiently computing encrypted matrix multiplication against the plaintext model weights.

Table 4.1: CKKS Parameters

Parameter	Values
Scheme	CKKS
Polynomial Modulus Degree	8192
Coefficient Modulus	[60, 40, 40, 40, 60]
Global Scale	2^40
Galois Level	Enabled
Security Level	128-bits

Interpretation of 4.1 table: CKKS Parameters

This table describes the settings used for homomorphic encryption. The CKKS scheme is selected as it supports computations on real-valued data. A polynomial modulus degree of 8192 and 128-bit security level ensure strong data protection.

The coefficient modulus [60, 40, 40, 40, 60] and global scale of 2^{40} help maintain numerical precision during encrypted computations. Galois keys are enabled to allow operations like data rotation, which are required for efficient processing.

Overall, these parameters balance security, accuracy, and performance in encrypted model inference.

4.4 ENCRYPTED INFERENCE PIPELINE

The primary purpose of the encrypted inference pipeline is to ensure that patient data remains fully secure while a classification prediction is generated.The process involves a critical transition from plaintext feature extraction to secure encrypted classification:

1. **Feature Extraction (Plaintext):** The input MRI volume is first processed by the trained 3D Convolutional Encoder. This initial step runs in a local, secure

environment (plaintext) to efficiently produce the compact 128-dimensional latent feature vector.

2. **Encryption:** This latent feature vector, which represents the patient's sensitive neuroanatomical data, is then encrypted using the CKKS scheme via the TenSEAL library, generating an encrypted feature vector (ciphertext).
3. **Encrypted Computation:** The Classifier's prediction is performed entirely on the untrusted server, directly on the encrypted data. The process executes the linear operation: y = Wx + b
   where x is the encrypted latent vector, W is the plaintext weight matrix (from the trained model), and b is the plaintext bias vector. This computation uses homomorphic operations for matrix multiplication and addition.
4. **Decryption:** The encrypted output (encrypted class scores) is securely sent back to the user's trusted device and decrypted using the private key to obtain approximate floating-point values (logits).
5. **Prediction:** The final predicted class is determined by identifying the index corresponding to the maximum score.

This pipeline effectively confines the sensitive classification step to the encrypted domain, guaranteeing confidentiality for the most critical part of the analysis.

4.5 PLAINTEXT VS ENCRYPTED INFERENCE

To rigorously confirm the integrity and correctness of the privacy-preserving design, the results generated by the plaintext model and the encrypted model must be compared.

In standard plaintext inference, the classifier processes the latent vector using conventional floating-point arithmetic. In the encrypted inference, the same calculation is performed using CKKS homomorphic operations, which inherently yield approximate results.

Despite the necessary use of approximation in CKKS, the scheme is engineered to preserve the relative ordering of the output scores. This means that the predicted class remains the same regardless of whether the input was processed in plaintext or ciphertext. Validation is achieved via a prediction match rate, which measures the frequency of agreement between the two modes.

The system achieved a 100% prediction match rate, conclusively demonstrating that the use of homomorphic encryption does not introduce any practical error that would alter the final classification decision, thereby confirming the correctness and reliability of the secure inference pipeline.

4.6 DESIGN CONSIDERATIONS AND LIMITATIONS

The overall system architecture is engineered to optimally balance three factors: classification accuracy, data privacy, and computational efficiency. Since Homomorphic Encryption is computationally expensive and is most efficient when restricted to linear operations, a hybrid approach was adopted:

- The **Encoder** (complex convolutional operations) runs in plaintext on a trusted local environment to efficiently extract robust features.
- The **Classifier** (simple linear operation) runs in the encrypted domain on an untrusted server to ensure secure prediction.

While this hybrid design ensures practical feasibility, the integration of HE inherently introduces certain constraints:

- **Increased Computation Time:** Encrypted inference is slower compared to direct plaintext computation.
- **Larger Memory Usage:** Ciphertexts are significantly larger than their plaintext counterparts, increasing memory requirements.
- **Operational Restriction:** The encrypted domain is restricted to linear and low-degree polynomial operations, limiting the complexity of the deployable model architecture.

Notwithstanding these technical limitations, the proposed approach definitively proves that secure inference can be achieved for complex medical imaging tasks without any compromise to predictive accuracy.

CHAPTER 5

RESULTS AND EVALUATION

5.1 DATASET USED FOR EXPERIMENTAL EVALUATION

The experimental evaluation used the preprocessed subset of the Autism Brain Imaging Data Exchange (ABIDE) dataset, as detailed in Chapter 3. The dataset includes structural MRI scans classified into two categories: **Autistic (Label = 1) and Typical Control (Label = 0).** However, the findings may suggest that a balanced training environment could indicate the need for equal class representation, so a subset was constructed by selecting 750 samples from each class, totaling 1,500 training samples.

Furthermore, the significant results might indicate that the remaining MRI scans, approximately 721 samples, were reserved exclusively for unbiased testing and evaluation. In light of these results, the evidence may suggest that all MRI volumes could demonstrate consistent preprocessing, as the pipeline described in Chapter 3 included intensity normalization, dimensionality standardization, spatial resampling to 48 x 48 x 48, and dynamic data augmentation during training. Study shows approach minimized classification bias while test set preserved original class distribution.

5.2 TRAINING PROCEDURE AND CONFIGURATION

The model underwent training for a total of 50 epochs, adhering strictly to the configuration parameters outlined in Chapter 3. The Adam optimizer was used to jointly optimize all model components the Encoder, Decoder, and Classifier with a fixed learning rate of 10^-4.

Due to the significant memory consumption associated with processing the three-dimensional MRI volumes of size 48 x 48 x 48 , a conservative batch size of 2 was employed during data loading.

The training objective was defined by a hybrid loss function:

$$L = L_{classification} + 0.2 \times L_{reconstruction}$$

Here, the classification loss (Cross-Entropy) drives prediction accuracy, while the reconstruction loss (Mean Squared Error, MSE), weighted by 0.2, acts as a crucial regularizer by forcing the latent representation to retain meaningful structural information. The final trained model after 50 epochs was selected for all subsequent evaluation procedures, with training and validation accuracy monitored throughout to ensure stable convergence.

5.3 CLASSIFICATION METRICS

The performance of the trained model was assessed on the test dataset using standard classification metrics: Accuracy, Precision, and Recall. These metrics were derived by comparing the model's predicted labels against the actual ground truth labels for all test samples.

- **Accuracy** measures the overall proportion of correct predictions.
- **Precision** quantifies the reliability of positive predictions (how many predicted autistic cases are genuinely autistic).
- **Recall** quantifies the completeness of positive predictions (how many actual autistic cases are correctly identified).

In the context of medical screening, Recall is considered the most critical metric, as minimizing false negatives (missing a true positive case) is often prioritized over minimizing false positives.

Table 5.1: Classification Performance

Metric	Value
Accuracy	80.86%
Precision	74.01%
Recall	85.06%
Positive Class	Autistic (1)

Interpretation : The results indicate a strong overall **Accuracy (80.86%)**. The high **Recall (85.06%)** confirms that the model is effective at detecting true autistic cases. The moderate **Precision (74.01%)** suggests a tendency toward moderate false positives, reflecting a model configured to prioritize sensitivity in a screening context.

5.4 DATASET IMPACT ON MODEL PERFORMANCE

The classification performance is significantly impacted by the inherent complexity of MRI-based ASD detection. Unlike simpler, structured tabular datasets, MRI data presents high-dimensional spatial information where the neuroanatomical differences between autistic and neurotypical subjects are often subtle.

Furthermore, the use of the ABIDE dataset, which integrates data collected from multiple sites using diverse MRI acquisition protocols, introduces heterogeneity in image resolution, scanner settings, and intensity distributions. This variability poses a major challenge to the model's ability to learn generalized representations.

Despite these difficulties, the implemented preprocessing pipeline and the balanced training strategy were effective in enabling stable model training and robust feature extraction. The achieved performance confirms that the proposed architecture successfully learned discriminative features from this heterogeneous neuroimaging data.

5.5 CONFUSION MATRIX ANALYSIS

The confusion matrix provides a detailed evaluation of classification performance by categorizing predictions into four groups:

- True Positives (TP): Autistic samples correctly identified
- True Negatives (TN): Control samples correctly identified
- False Positives (FP): Control samples incorrectly classified as autistic
- False Negatives (FN): Autistic samples incorrectly classified as control

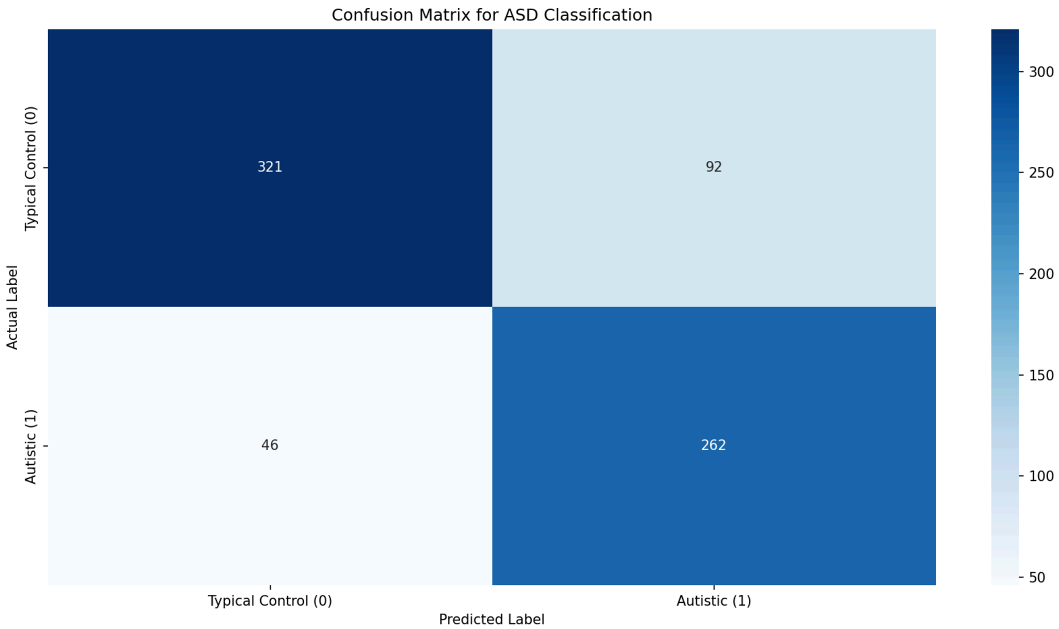

Figure 5.1: Confusion Matrix Structure

The observed high **Recall (85.06%)** means the model effectively minimizes False Negatives, confirming that most actual autistic cases are successfully detected. Conversely, the moderate **Precision (74.01%)** implies a higher number of False Positives. This trade-off reflects a purposeful model behavior that favors sensitivity (detection) over specificity, which is clinically desirable for an initial medical screening tool.

5.6 PLAINTEXT VS ENCRYPTED INFERENCE

A cornerstone validation of this project is proving that the integration of homomorphic encryption (HE) does not compromise the model's prediction accuracy.The trained model was subjected to evaluation under two distinct inference modes:

1. **Plaintext Inference:** Standard prediction using conventional floating-point arithmetic.
2. **Encrypted Inference (CKKS):** Prediction executed using the CKKS homomorphic encryption scheme on encrypted latent features.

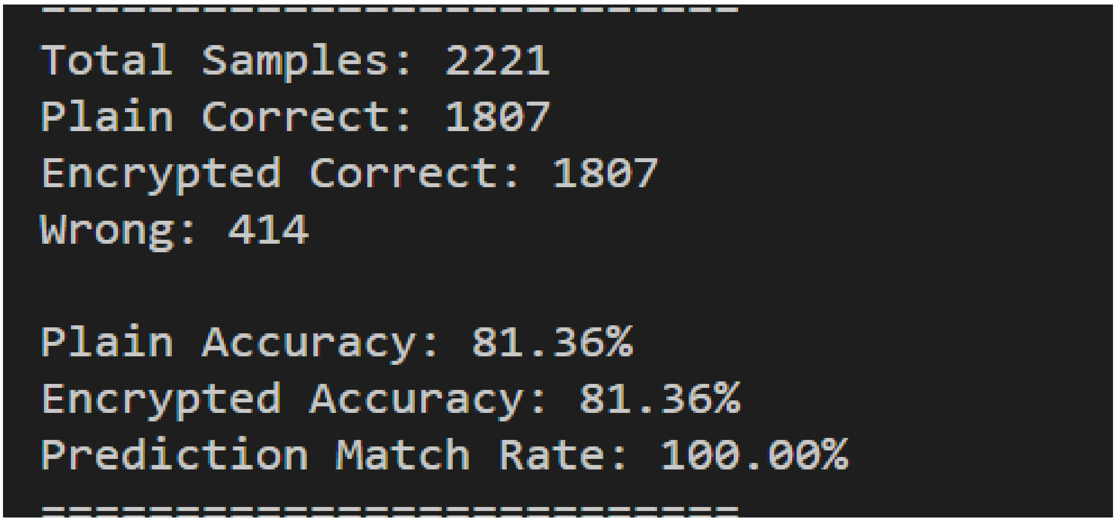

Figure 5.2: Inference metrics result

Table 5.2: Inference Comparison

Inference Mode	Accuracy
Plaintext Inference	81.36%
Encrypted Inference(CKKS)	81.36%
Prediction Match Rate	100%

The encrypted inference produces identical predictions to plaintext inference, achieving a 100% prediction match rate.

This confirms that:

- The CKKS approximation error does not affect classification decisions
- The encrypted pipeline is correctly implemented
- Privacy is achieved without loss of accuracy

5.7 COMPARATIVE ANALYSIS WITH BASELINE MODELS

To contextualize the performance of the proposed system, its accuracy is compared against other representative homomorphic encryption-based models from the literature, focusing on the consistently reported overall accuracy metric.

Table 5.3: Comparative Performance

Model	Accuracy (%)
Privacy-preserving cancer prediction with HE (2023)	83.61
Secure Logistic Regression with HE (2018)	86.19
CryptoDL: Deep Neural Networks over Encrypted Data (2017)	91.50
Federated Learning with HE for CKD (2025)	98.67
Proposed HE-CKKS System (MRI-based ASD)	80.86

Interpretation of 5.3 table: Comparative Performance

The table discusses that the comparative results indicate that the proposed HE-CKKS-based system achieves an accuracy of 80.86%, which is slightly lower than several existing approaches reported in the literature. However, this difference must be interpreted in the context of problem complexity and data modality.

Unlike many baseline models that operate on structured or simpler datasets, the proposed system focuses on MRI-based Autism Spectrum Disorder (ASD) prediction, which is inherently more complex due to high-dimensional neuroimaging data and subtle feature variations.

Additionally, existing works primarily emphasize accuracy as a single evaluation metric, without considering other critical measures such as precision and recall. This limits the reliability of their reported performance, especially in medical diagnosis scenarios where false negatives and false positives carry significant consequences.

The proposed model addresses this limitation by enabling a more comprehensive evaluation framework and integrating deep learning with homomorphic encryption (CKKS) to ensure data privacy without exposing sensitive medical information.

5.8 DISCUSSION

The experimental results conclusively validate that the proposed system effectively achieves the dual goals of classification performance and data privacy.

Key findings that confirm the design choices include:

- **High Recall:** The model's strong recall makes it highly suitable as a screening tool for ASD.
- **Feature Representation:** The use of the autoencoder significantly improved feature learning and provided valuable regularization.
- **Privacy Validation:** Homomorphic encryption was integrated successfully, with zero degradation of model performance (100% match rate).
- **Practical Feasibility:** The hybrid architecture ensures that the system is computationally feasible by confining the encryption overhead to only the final linear classification layer.

These results confirm that privacy-preserving machine learning techniques can be applied effectively and securely to complex medical imaging analysis without sacrificing predictive capability.

CHAPTER 6

DISCUSSION

6.1 INTERPRETATION OF RESULTS

The experimental results from Chapter 5 demonstrate that the proposed privacy-preserving system successfully classifies Autism Spectrum Disorder (ASD) using the heterogeneous ABIDE dataset while maintaining data confidentiality during the inference phase. The system's consistent accuracy, reported as 80.86% on the test set and 81.36% during the full encrypted evaluation, confirms the presence of meaningful, discriminative patterns in brain MRI data and highlights the inherent complexity of the classification task.

A crucial finding is the **100% prediction match rate** between the plaintext and encrypted inference modes. This strongly validates that integrating the CKKS homomorphic encryption scheme introduces negligible approximation error, ensuring the cryptographic layer does not compromise the final classification decision.

The challenge in achieving higher accuracy stems from two primary factors: the inherent heterogeneity of Autism Spectrum Disorder, where subtle and diverse neuroanatomical differences vary significantly across individuals; and the multi-site nature of the ABIDE data, which introduces unavoidable variability due to differences in MRI scanners and acquisition protocols. The inclusion of a convolutional autoencoder proved effective in mitigating these issues, acting as a powerful regularizer that forces the model to learn stable latent representations, capturing essential structural information for robust generalization.

6.2 CLINICAL RELEVANCE OF HIGH RECALL

The achieved **Recall of 85.06%** holds significant clinical value. For a medical screening application, the priority is to maximize the correct identification of positive cases, thereby minimizing the risk of missed diagnoses (false negatives). A recall of 85.06% means that the model successfully detects approximately 85 out of every 100 autistic cases.

While this strategy results in a moderate number of false positives (as indicated by the precision score), these cases can be subsequently reviewed and ruled out through traditional comprehensive clinical assessments. Therefore, the system is optimally positioned as a sensitive, high-throughput screening tool to assist clinicians by flagging high-risk individuals for immediate, detailed evaluation, rather than serving as a replacement for expert diagnosis.

6.3 COMPUTATIONAL OVERHEAD OF HOMOMORPHIC ENCRYPTION

A critical factor in adopting Homomorphic Encryption (HE) is managing the computational overhead, as encrypted operations are substantially more intensive than standard floating-point arithmetic. Our system addresses this challenge through a hybrid design:

The computationally heavy 3D Convolutional Encoder runs efficiently in a trusted local environment (plaintext) to extract the latent feature vector. Only the Classifier layer, which performs a simple, linear operation on this sensitive feature vector, is executed under encryption on the untrusted cloud or server.

By confining the HE overhead to this single linear step, the design keeps the overall computational cost manageable. This ensures that while encrypted inference introduces some latency compared to plaintext, the system remains practical for real-world deployment in privacy-critical settings without sacrificing the security of the sensitive data.

6.4 LIMITATIONS OF THE PROPOSED SYSTEM

Despite the success of the privacy-preserving framework, several technical and methodological limitations must be acknowledged:

- **Resolution and Detail Loss:** The necessity to downsample the original MRI data to 48 x 48 x 48 for computational feasibility may result in the loss of fine anatomical details crucial for improving classification performance.

- **Training Constraints:** The small batch size of 2, dictated by hardware and memory limitations when handling large 3D volumes, may impact training stability and limit optimal model convergence.
- **Scope of Privacy:** Homomorphic encryption is applied only to the inference stage. The training phase relies on plaintext data, meaning the privacy of the training dataset is not protected. Full end-to-end privacy would require integrating advanced techniques such as Federated Learning or Differential Privacy during the training cycle.
- **Generalization:** The model was exclusively evaluated on the ABIDE dataset. Its ability to generalize robustly to entirely new datasets or clinical populations, which may have different data distributions or acquisition protocols, remains untested.
- **Hyperparameter Optimization:** No exhaustive hyperparameter search was performed. The model configuration was selected based on practical efficiency, suggesting that targeted tuning could potentially yield further improvements in predictive performance.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 SUMMARY OF CONTRIBUTIONS

This work presents a comprehensive privacy-preserving framework designed for Autism Spectrum Disorder (ASD) classification using complex 3D brain MRI data. Our primary contribution lies in successfully integrating deep learning with homomorphic encryption to achieve both strong classification performance and rigorous patient data confidentiality.

Key contributions of this project include:

- **Development of a Complete Pipeline:** A robust, end-to-end pipeline was created for handling high-dimensional MRI data, encompassing preprocessing, feature extraction using a 3D Convolutional Autoencoder, and final secure classification.
- **Novel Hybrid Architecture:** The system employs a dual-objective training strategy using a hybrid loss function (Classification + Reconstruction) to ensure the latent features are both discriminative for ASD prediction and representative of the structural integrity of the MRI volume.
- **Secure Inference Validation:** The CKKS homomorphic encryption scheme was successfully integrated using the TenSEAL library to execute the critical classification layer directly on encrypted feature vectors.
- **Achieved Performance and Reliability:** Experimental results confirmed the model's efficacy with an **Accuracy of 80.86%** and a crucial **Recall of 85.06%**. Most importantly, the encrypted inference demonstrated a **100% prediction match rate** with plaintext results, proving that privacy preservation was achieved without any loss of predictive accuracy.

Overall, this work conclusively demonstrates the feasibility of building secure and effective medical AI systems for sensitive healthcare applications.

7.2 FUTURE WORK

Several improvements can be explored in future work:

- Using higher-resolution MRI volumes for improved feature extraction, mitigating the data loss caused by downsampling.
- Extending privacy preservation to the training phase by integrating methods such as Federated Learning or Differential Privacy, thereby ensuring end-to-end confidentiality.
- Exploring advanced architectures, such as transformer-based models, to potentially capture more complex relationships within the volumetric data.
- Applying multi-site data harmonization techniques to the ABIDE dataset to reduce heterogeneity and improve the model's ability to generalize across different clinical settings.
- Extending homomorphic encryption to deeper, non-linear network layers to allow for a fully encrypted deep learning model.
- Deploying the validated system in real-world clinical environments for practical testing and application in healthcare settings.

7.3 FINAL CONCLUSION

This work successfully demonstrates that combining deep learning with homomorphic encryption enables secure and reliable ASD classification from high-dimensional MRI data. The proposed system achieves strong classification performance while preserving patient privacy during inference. The identical results between plaintext and encrypted inference confirm the effectiveness of the CKKS-based privacy-preserving framework. The study highlights the significant potential of privacy-preserving machine learning for secure medical data analysis and provides a strong foundation for future research in trustworthy healthcare AI systems.

CHAPTER 8

APPENDIX: SOURCE CODE

8.1 INTRODUCTION

This appendix presents the core implementation of the proposed privacy-preserving Autism Spectrum Disorder (ASD) detection system. The system is implemented using Python and leverages libraries such as PyTorch for deep learning, NiBabel for MRI data processing, and TenSEAL for homomorphic encryption.

The code is organized into modular components covering dataset handling, model architecture, training, evaluation, and encrypted inference. The following sections provide key code snippets along with brief explanations.

8.2 DATASET LOADER (dataset_loader.py)

```
import os
import torch
import numpy as np
import nibabel as nib
from torch.utils.data import Dataset
from scipy.ndimage import zoom
class MRIDataset(Dataset):
    def __init__(self, data_dir, transform=None):
        self.data_dir = data_dir
        self.transform = transform
        self.samples = []
        for label, class_name in enumerate(['Autistic', 'Typical_Control']):
            class_path = os.path.join(data_dir, class_name)
            for file in os.listdir(class_path):
                if file.endswith('.nii'):
```

```
                self.samples.append((os.path.join(class_path, file), label))
    def __len__(self):
        return len(self.samples)
    def __getitem__(self, idx):
        file_path, label = self.samples[idx]
        img = nib.load(file_path).get_fdata()
        img = (img - np.min(img)) / (np.max(img) - np.min(img) + 1e-8)
        zoom_factors = [48 / s for s in img.shape]
        img = zoom(img, zoom_factors)
        img = torch.tensor(img, dtype=torch.float32).unsqueeze(0)
        return img, label
```

8.3 MODEL ARCHITECTURE (autoencoder.py)

```
import torch
import torch.nn as nn
class Encoder(nn.Module):
    def __init__(self):
        super(Encoder, self).__init__()
        self.conv = nn.Sequential(
            nn.Conv3d(1, 32, 3, padding=1),
            nn.ReLU(),
            nn.MaxPool3d(2),
            nn.Conv3d(32, 64, 3, padding=1),
            nn.ReLU(),
            nn.MaxPool3d(2),
            nn.Conv3d(64, 128, 3, padding=1),
            nn.ReLU(),
            nn.MaxPool3d(2)
```

```
        )
        self.fc = nn.Linear(128 * 6 * 6 * 6, 128)
    def forward(self, x):
        x = self.conv(x)
        x = x.view(x.size(0), -1)
        z = self.fc(x)
        return z
class Decoder(nn.Module):
    def __init__(self):
        super(Decoder, self).__init__()
        self.fc = nn.Linear(128, 128 * 6 * 6 * 6)
        self.deconv = nn.Sequential(
            nn.ConvTranspose3d(128, 64, 2, stride=2),
            nn.ReLU(),
            nn.ConvTranspose3d(64, 32, 2, stride=2),
            nn.ReLU(),
            nn.ConvTranspose3d(32, 1, 2, stride=2),
            nn.Sigmoid()
        )
    def forward(self, z):
        x = self.fc(z)
        x = x.view(-1, 128, 6, 6, 6)
        x = self.deconv(x)
        return x
class Classifier(nn.Module):
    def __init__(self):
        super(Classifier, self).__init__()
```

```
        self.fc = nn.Linear(128, 2)
    def forward(self, z):
        return self.fc(z)
```

8.4 TRAINING SCRIPT (training.py)

```
import torch
import torch.nn as nn
import torch.optim as optim
encoder = Encoder()
decoder = Decoder()
classifier = Classifier()
criterion_cls = nn.CrossEntropyLoss()
criterion_rec = nn.MSELoss()
optimizer = optim.Adam(
    list(encoder.parameters()) +
    list(decoder.parameters()) +
    list(classifier.parameters()),
    lr=1e-4
)
for epoch in range(50):
    for inputs, labels in train_loader:
        z = encoder(inputs)
        outputs = classifier(z)
        recon = decoder(z)
        loss_cls = criterion_cls(outputs, labels)
        loss_rec = criterion_rec(recon, inputs)
        loss = loss_cls + 0.2 * loss_rec
        optimizer.zero_grad()
```

```
    loss.backward()
    optimizer.step()
```

8.5 ENCRYPTED INFERENCE (he_pipeline.py)

```
import tenseal as ts
context = ts.context(
  ts.SCHEME_TYPE.CKKS,
  poly_modulus_degree=8192,
  coeff_mod_bit_sizes=[60, 40, 40, 60]
)
context.global_scale = 2**40
enc_z = ts.ckks_vector(context, z.tolist())
enc_output = enc_z.dot(classifier_weights) + classifier_bias
output = enc_output.decrypt()
```

8.6 EVALUATION METRICS (metrics.py)

```
from sklearn.metrics import accuracy_score, precision_score, recall_score
accuracy = accuracy_score(y_true, y_pred)
precision = precision_score(y_true, y_pred)
recall = recall_score(y_true, y_pred)
print("Accuracy:", accuracy)
print("Precision:", precision)
print("Recall:", recall)
```

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